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Review

Concrete and steel bridge Structural Health Monitoring—Insight into choices for machine learning applications

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ABSTRACT

Structural Health Monitoring (SHM) systems have been installed on bridges across the world at an increasing rate in recent years, providing vital data for bridge assessment and maintenance. Machine Learning (ML) is efficient in data analyses such as classification and regression, and capable of improving its accuracy by learning from data without the need for step-to-step programming. The implementation of ML methods in bridge SHM studies has become more popular in recent years for its ability to detect damages on concrete and steel caused by material deterioration and to perform condition assessment on bridge structures. There have been several review articles discussing ML applications in SHM which mostly provide broad discussions across different civil engineering structures. In this article, different ML applications in the bridge SHM study are summarised and discussed. Detailed critiques of each types of ML applications are provided. Finally, recommendations are made for the future study of ML applications in bridge SHM to fill the current research gaps.

1. Introduction

Long-span bridges have been playing an important role in human society. These bridges provide important connections across rivers, valleys, bays and other natural formations. Therefore, bridges normally have a relatively long design life span, e.g. 120 years according to the Eurocode (EN1990.2002) [1]. However, it has been discovered that many of these bridges do not live up to their designed life span and have premature failures. For example, in 2021, 7.5% of bridges which carry over 178 million trips in the US are considered structurally deficient and 42% of bridges in the USA are at least 50 years old [2]. In the UK, the Forth Road Bridge spanning the Firth of Forth, was found with truss end link failure in December 2015. Premature failures on long-span bridges are mainly caused by: discrepancies between the designed and constructed structures [3] and the unexpected increase in the traffic [3], eventually resulting in material deteriorations. Traditionally, bridge assessments are done by visual inspections to identify damages. However, it was found that visual inspections lack accuracy due to the subjectivity of the inspectors [4] and can be expensive and time-consuming [5]. Therefore, the continuous monitoring of in-service bridges is in high demand in order to detect changes and damages. Inspired by the rapid development of the IT industry, the implementation of the Structural Health Monitoring (SHM) system has become extremely popular in recent years. SHM can be defined as a complex system consisting of sensory, data manipulation, modelling and damage detection [6]. SHM has widely been adopted in the mechanical [7], aeronautics [8] and civil engineering fields [9–11]. For SHM systems on large civil infrastructures, huge amount of continuous monitoring data can be generated daily. Therefore, research on bridge SHM systems has become more popular recently.

There are generally two approaches to studying the SHM data, model-based and data-based. The model-based methods usually require a precise finite element (FE) model of the monitored structure. Data collected from the SHM system will be applied to the digital replica of the structure. Then, the changes in the structure can be inspected and defects will be identified by comparing the different states of the FE model. There have been a variety of researches using model-based methods in the past decades [12–16]. However, these methods are usually applied to experimental structures and small bridges. Their

Abbreviations: AI, Artificial intelligence; BRNN, Bidirectional recurrent neural network; CNN, convolutional neural network; DNN, Deep neural network; FE, Finite element; LSTM, Long short-term memory; ML, Machine learning; MLP, Multilayer perceptron; PCA, Principal component analysis; RNN, Recurrent neural network; SHM, Structural health monitoring; SVM, Support vector machine; UAVs, Unmanned aerial vehicles

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implementation can be difficult for long-span bridges due to the complexity of the structures. Developing and analysing their FE models will be extremely time-consuming or even impractical. On the other hand, the normal operational status of the structure is required to identify any discrepancies caused by defects. Therefore, this has led to an increasing interest in data-based methods where analysis is carried out purely based on the monitoring data with statistical models and the digital replica of the bridges is not needed. Traditional statistics models have been used to analyse SHM data in various researches. There are a few disadvantages to these conventional models. First, step-by-step calculations are needed for these methods, which can complicate the calculation procedures. Secondly, the SHM system on large infrastructures generates a massive amount of data in multiple channels. The complexity of the datasets weakens the generalisation of the statistical models, reducing the reliability of the results. With the development of computer hardware and new algorithms, Machine Learning (ML), proven efficient with complex datasets [17], is gradually deployed to SHM problems.

Machine learning can be defined as computer algorithms enabling programs to improve with experience automatically using data. [18] ML has achieved great success in multiple fields, e.g. computer vision [19], natural language processing [20] and data mining [21,22]. Machine learning application in the civil engineering field is fairly recent. For instance, Farrar and Worden [23] introduced the theoretical background of different machine learning algorithms and their potential applications in SHM data analysis but lacks case studies in large civil infrastructures.

This paper provides an overview of different machine learning applications in data analysis of in-service bridge SHM systems worldwide in recent years. The scope of this paper is as follow:

- Briefly introduce the bridge structural health monitoring system implemented across the world
- Provide the necessary background on different machine learning methods used for SHM studies
- Present various research developments, discuss and compare the highlights and drawbacks
- Summarise the suitability of different machine learning methods for different SHM problems
- Finally discuss the future trends

Compared with other review articles in the similar topic, this paper not only presents applications of different machine learning methods but also the necessary data preprocessing steps for different data. Furthermore, a clear and concise summary table is given for applications of each machine learning method. The overall structure of this article is as follows: Section 2 introduces the implementation of SHM systems on long-span bridges. Section 3 presents the machine learning concepts and several popular machine learning algorithms. Section 4 shows different applications of machine learning methods in the bridge SHM studies.

2. Bridge SHM systems

Several bridge components are commonly subjected to premature failures, for example, the expansion joints, due to the repetitive influence of vehicle loads [24]; the bridge decks, due to freeze-thaw action, alkali silica reaction and reinforcement corrosion [25]. Furthermore, suspension and cable stayed bridges are at risk of cable degradation. Apart from bridge degradations, other factors also affect the operations of bridges, such as ice accretion on cables, endangering the public. Therefore, a continuous monitoring system installed on long-span bridges is needed to identify these hazards. Structural health monitoring systems are installed on several components of bridges for various purposes, including confirming design, construction aid, demonstration and life extension [26]. Furthermore, there is a tendency that more SHM systems on bridges are designed for life extension purpose, allowing continuous monitoring during the operation stage. Table 1 shows several examples of SHM system installation on long-span bridges worldwide.

An SHM system installed on long-span bridges consists of the following components: a sensory system, local or cloud storage facilities and finally, a real-time assessment process, alerting the bridge management team if anomalies are observed. The sensory system contains sensors that measure loading (wind, traffic), structural responses (acceleration, strain, displacement) and environmental factors (temperature) [34,35]. Regarding sensor types, most SHM systems are based on single-point sensors such as strain gauges, accelerometers and GPS-based sensors to monitor displacement [36]. However, in recent years, fibre-optic sensors are gaining popularity for their higher accuracy, lightweight, stability and low maintenance cost [37] - however, the installation cost may be high. Fig. 1 shows a few examples of sensors installed on the Queensferry Crossing, UK. The Queensferry Crossing is currently monitored by 2184 sensors [33].

The monitored data will then be simply processed before they are stored, e.g. downsampling to save storage space. The real-assessment process will further investigate any anomalies in the data. Normally, these systems allow the bridge management team to visualise real-time monitoring data from any sensors. These anomaly detections are generally set to trigger by a preset threshold for each sensor of interest.

3. Machine learning

This section presents a brief introduction of the fundamental concepts of machine learning and the algorithms of several ML models.

Machine learning is a type of Artificial intelligence (AI) where models are trained with sample data (input) to make predictions or decisions without human intervention during the training stage. It can be concluded that there are 3 parts in the procedure, input data, ML model training and output data. Depending on different problems, the selection of the machine learning model and the input data preparation can vary.

3.1. Input data

3.1.1. Data cleaning

SHM systems collect a great number of different types of data, e.g. time-series data (temperature, strain, deflection), discrete data (vehicle passing from the weigh-in-motion system) and sometimes imagery data (concrete surface photos). However, the dimension of the input data to an algorithm must maintain consistency and also the raw data from SHM systems. Furthermore, sensors collect data at different sampling rates even under the same SHM system on one bridge. Therefore, the raw data cannot be used as input to machine learning algorithms. Data manipulation that reshapes the dimension of the raw data is essential.

After manipulating the data, the next step is to ensure the data quality which is essential to any machine learning task. Poor quality data can lead to catastrophic consequences and even cause projects to fail [38]. In the SHM data, several common types of data errors are listed below.

- **Missing data** This is presented as gaps in a continuous dataset. Normally, missing data is caused by malfunctioning sensors caused by power issues or errors when certain data points are recorded in the storage system. In the raw data, missing parts are shown as empty entry or *Nan*, which the algorithms will treat as errors and leads to failed training. Therefore, these missing parts must be imputed.

Table 1
Examples of SHM implementations on long-span bridges.

Bridge	Bridge type	Total length (m)	SHM installation time	Number of sensors	Sensors	Reference
Golden State Bridge California, USA	Suspension	1966	1993	64	Accelerometers	Çelebi [27]
Tsing-ma Bridge Hong Kong	Suspension	2160	1997	280	Environmental (wind, temperature) Response (GPS, acceleration, strain)	Li et al. [28]
Tamar Suspension Bridge Plymouth, UK	Suspension	642	2001	Around 100	Environmental (wind, temperature) Response (cable tension, deck level)	Koo et al. [29]
Humber Bridge Yorkshire, UK	Suspension	2220	2010–2012	12	Environmental (wind, temperature) Response (GPS, Accelerations)	Brownjohn et al. [30]
Sutong Bridge Jiangsu, China	Cable-stayed	8206	2008	1078	Environmental (wind, temperature) Response (GPS, Accelerations, cable tension, deck level, etc.)	Wang et al. [31]
Forth Road Bridge Edinburgh, UK	Suspension	2517	2015	192	Environmental (wind, temperature) Response (GPS, Accelerations, strain, etc.) Operational (Weigh-in-motion)	Xu et al. [32]
Queensferry Crossing Edinburgh, UK	Cable-stayed	2700	2018	2184	Environmental (wind, temperature) Response (GPS, Accelerations, strain, etc.) Operational (Weigh-in-motion)	Cousins et al. [33]

- **Outliers** Outliers are data that significantly differ from other records. They often indicate a measurement error by sensors affected by environmental factors. These outliers usually exceed the reasonable range of its data type, e.g. 100 °C for air temperature. These data must be excluded from the dataset as it will greatly reduce the accuracy of the algorithms since the variation of the data will be flattened with the existence of extreme values. On the other hand, some outliers are within the reasonable range, indicating abnormal events. These outliers should be preserved. Separate classification algorithms should be used to distinguish outliers from the raw data. For example, the *k*-nearest neighbours classifier (*k*-NN) can be used to identify outliers based on distance [39]. AutoRegressive Integrated Moving Average (ARIMA) model [40] and Isolation Forest [41] are also observed to be used to detect outliers for time series data.

Proactive measures can be implemented to safeguard data integrity during the data collection stage. In most Structural Health Monitoring (SHM) systems, redundancy is introduced at the sensor installation phase. For instance, multiple strain gauges are affixed to the same section of the main truss end post on the Forth Road Bridge, as depicted in Fig. 2. Consequently, if a specific sensor experiences data loss, data from a neighbouring sensor of the same type can be utilised as a substitute.

3.1.2. Data normalisation

Normalisation is necessary to further improve the data quality after the data cleaning process. Different sensors in an SHM system have different measurement scales, e.g. strain, displacement and temperature. These measurements show significantly different data ranges from each other. Data normalisation is a process to adjust each measurement to the same scale so each feature will have the same numerical contribution to the algorithms [42]. One of the most commonly used normalisation methods is the min–max normalisation, where the data are rescaled to $[0, 1]$. The general formula can be written as follows:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

Other data normalisation methods are also used, such as z-score normalisation where the normalised data has a mean of 0 and a standard deviation of 1, shown in Eq. (2); MaxAbsScaler which scales and translates each feature individually such that the maximal absolute value of each feature in the training set will be 1, shown in Eq. (3).

$$X' = \frac{X - \bar{X}}{\sigma} \tag{2}$$

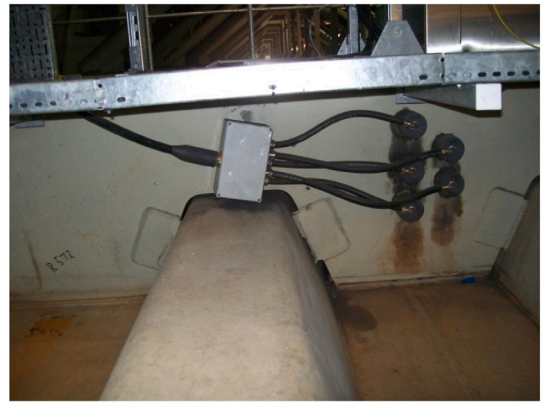
where X' is the scaled value, X is the original value, $\bar{X}$ is mean of the dataset and σ is the standard deviation.

$$X' = \frac{X}{\max(|X|)} \tag{3}$$

3.1.3. Feature engineering

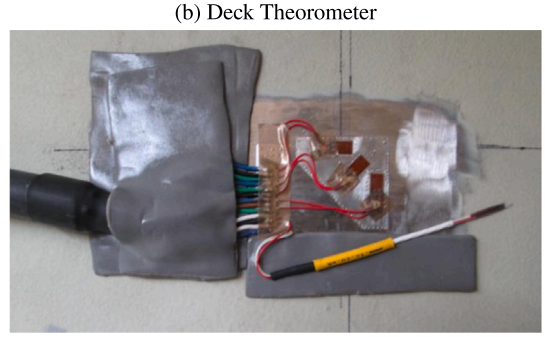
Finally, a step called feature engineering is applied to the data. Feature engineering can be described as a process to extract features, e.g. characteristics, from the raw data. It will either add extra features based on the raw measurements, for instance, one-hot encoding for categorical data, an example of which is depicted in Table 2 with four

(a) Rainfall Senesor(height: 39cm, diameter: 18.5cm)



(b) Deck Theorometer

(c) Displacement Transducer ($6cm \times 4cm$)



(d) Dynamic Strain Gauge ($80cm \times 60cm \times 45cm$ with enclosure)

Fig. 1. Sensors on queensferry crossing.

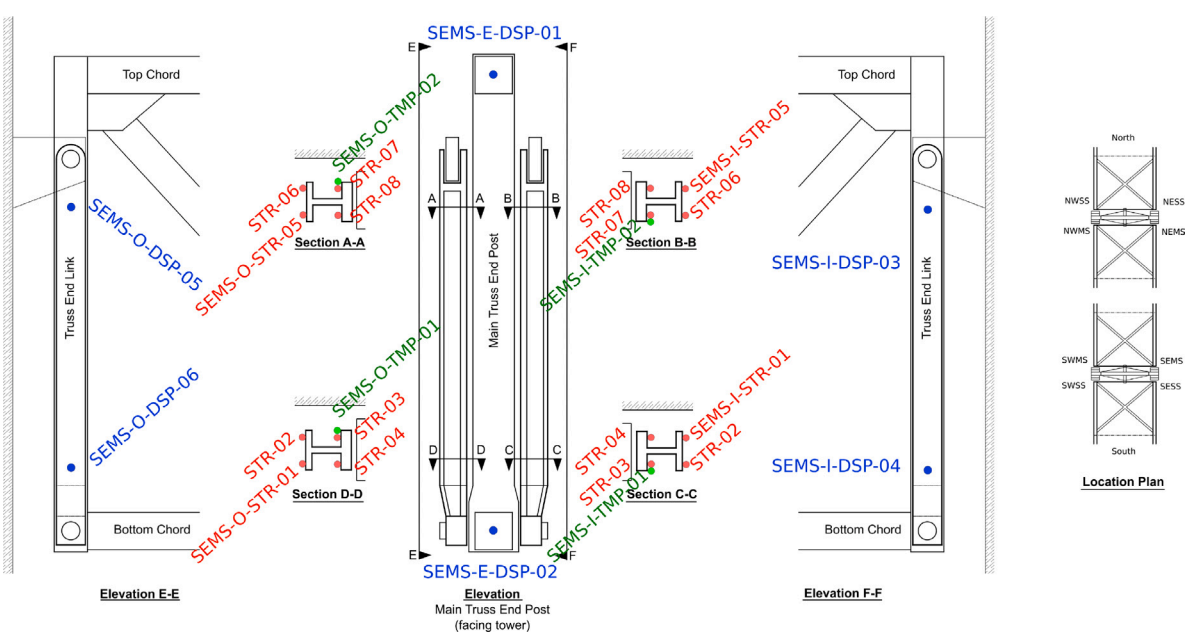
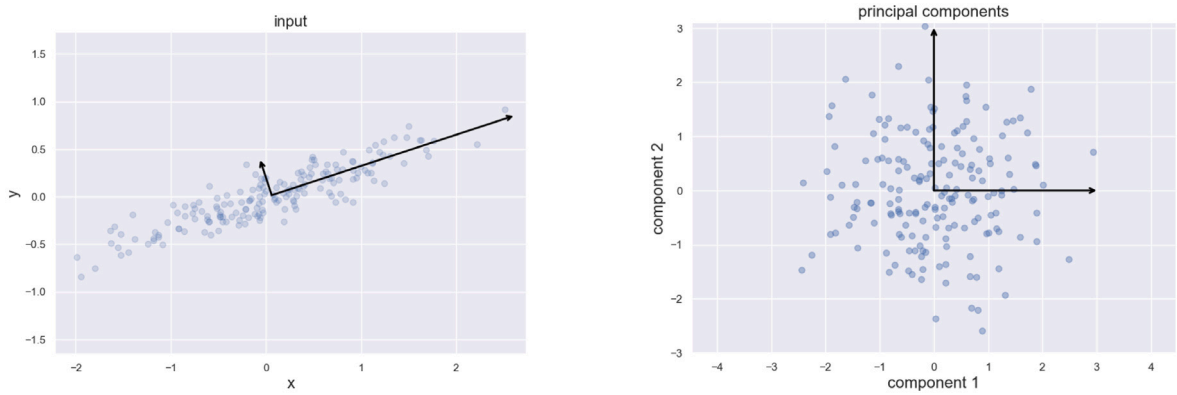


Fig. 2. Sensors layout on the forth road bridge.



(a) Original data (b) Principal components

Fig. 3. PCA example.

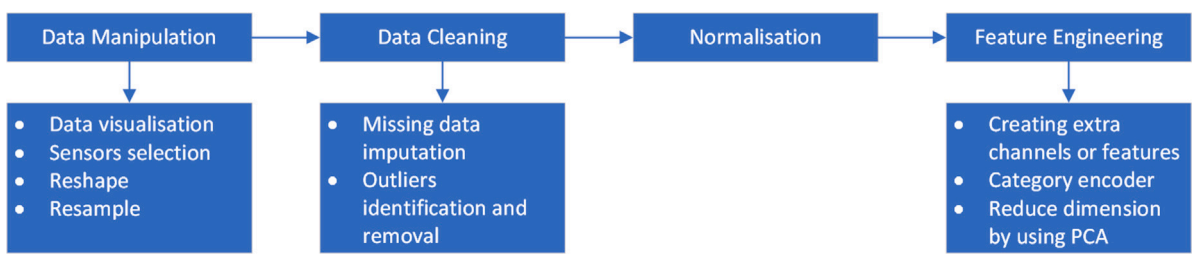


Fig. 4. Data preparation.

Table 2
One-hot encoding.

Undamaged	1	0	0	0
Minor	0	1	0	0
Major	0	0	1	0
Critical	0	0	0	1

classes utilising one-hot encoding. Another example is to reduce the dimension of the raw dataset by implementing Principal Component Analysis (PCA), an example of which is illustrated in Figs. 3(a) and 3(b). Dong and Liu [43] described a variety of feature engineering approaches for different data types. In general, feature engineering strategies depend highly on the data and algorithms. There are no universal procedures. Efficient feature engineering will contribute significantly to the generalisation of the machine learning models but it is also labour-intensive [44].

In general, the data preprocessing to prepare the input data for the subsequent machine learning training involves dimension reshaping, missing data imputation, outliers handling, normalisation and feature engineering etc. It can be summarised in Fig. 4. The quality of raw SHM data varies. Due to their environmental conditions, the stability of sensors is normally severely affected for long-span bridges and their measurements can show very poor quality. Therefore, data preprocessing is an indispensable procedure to increase the accuracy of the algorithm even though it is generally much more time-consuming than the machine learning training process.

3.2. Algorithms

In this section, a brief introduction is given for the fundamental concepts of machine learning and the algorithms of several ML models are presented.

As defined in Section 1, ML models will improve their accuracy by automatically learning from the existing data. Therefore, any ML algorithm has three essential parts: input data, the ML model and the output. The learning process is often referred to as "training".

Many different kinds of data can serve as input, for example, images, texts, time-series data like SHM monitoring data etc. Regardless of their different types, data pre-processing involves manipulation, dropping and sometimes imputation before the data can be used in the ML model. This step is essential as it will not only ensure the format of different input data is consistent for the ML models but also improve the overall quality of the data. If a large amount of noise or irrelevant information is present in the inputs, the ML models will be misled and the output is usually meaningless. There are a few commonly used methods in data pre-processing, including outliers removal, normalisation, encoding and feature extraction etc.

The second part is different ML models. For SHM data analysis and some other applications, the ML approaches can be classified into 3 categories, supervised learning, unsupervised learning and semi-supervised learning:

- **Supervised learning**: The desired outputs are already prepared in the training set, often referred to as the "labelled data". They will "guide" the training process. Therefore, the core of supervised learning algorithms is to minimise the discrepancies between the training output and the labelled data. A supervised learning model can make two kinds of predictions: classification and regression. In an SHM problem, classification can be the determination of the structure status, e.g. normal or critical; regression can be the prediction of the structural responses, e.g. deflections.
- **Unsupervised learning**: In an unsupervised learning model, the input set lacks the "labelled data". This is normally because the labels are unknown at the data collection stage. Therefore, the core of unsupervised learning is to identify the commonalities among the data and put them into different clusters.
- **Semi-supervised learning**: These models train with both labelled and unlabelled data. Typically, the amount of labelled data is

limited, while most are unlabelled. A semi-supervised learning algorithm will be guided with labelled data and the unlabelled are used to uncover hidden structures in the data [45]. Within context of SHM, a small amount of damage scenarios will be labelled to train a classifier. Then, the classifier is used to predict on unlabelled data and create labels. The most confident unlabelled instances and their predicted labels are then added to the training set, and the classifier is retrained.

Some of the commonly used machine learning algorithms in an SHM study are presented below. SVM is a supervised learning model for binary classification. SVM was first proposed by Cortes and Vapnik [46]. Apart from linear classification, SVM is also capable of non-linear classification using kernel functions.

For regression problems, decision tree is one of the popular models. A decision tree is a tree-like structure consisting of decision nodes, chance nodes and end nodes. This model starts with a decision node where a decision is made. Then in a chance node, multiple uncertain intermediate outcomes are presented. Finally, it reaches the end nodes, showing the outcomes. Decision tree is relatively easy to understand. Multiple decision trees can be constructed to improve its accuracy to form the random forest model, created by Ho [47].

Artificial neural networks (ANNs) have received the most attention and development compared to other ML models in recent years. There are many types of ANNs developed by researchers which have shown great success in complicated prediction tasks. Fig. 5 presents the structure of a very basic ANN model, the multilayer perceptron (MLP). There can be multiple hidden layers in the model and each hidden layer contains multiple neurons which use a non-linear activation function, e.g. sigmoid. Therefore, the function represented by the neuron can be written as:

$$a = f(z) = f(\sum_{j=1}^{m} x_j w_j + w_0) \tag{4}$$

where x is the input vector, w_j and w_0 are the weights, f is the non-linear function and a is the output vector of the neuron. The output vector of the neuron can also be the input vector for the neurons in the next hidden layer. During training, MLP is repeatedly presented with the training data set. All the weights in the network are adjusted using a technique called *Backpropagation* [48]. The detailed mathematical procedures of the backpropagation is given by Bishop et al. [49]. The basic idea is to calculate the gradient of the loss, discrepancies between the desired outputs and the MLP outputs, with respect to the weights and then to adjust the weights in the direction of the steepest gradient. Backpropagation is the most important concept in ANNs.

Based on the idea of MLP, many different types of ANNs are proposed and proven feasible and more efficient in a certain specialised field. For example, Convolutional Neural Network (CNN), used in deep learning (DL), is popular in image processing tasks. Unlike MLP, a CNN incorporates additional convolutional layers prior to the fully connected hidden layers. These convolutional layers utilise a set of learnable filters to scan the input data and identify salient features. Deep learning is a variety of ANNs with multiple levels of representation, enabling complex functions and relations to be learnt as each layer in the deep CNN model is capable of identifying a certain type of motif [50]. However, the complexity of the deep learning models is a great challenge for computer hardware to process. Therefore, to address this issue, Chellapilla et al. [51] introduced the use of graphics processing units (GPUs) for deep learning training, which is over 3x faster than the conventional hardware . Furthermore, Cireşan et al. [52] proves GPUs are even more than 40 times faster than standard microprocessors training multilayer perceptron. As a result, the implementation of complex multi-layer deep neural networks has become possible with the development of GPU-based neural network training. There have been several successful deep neural network (DNN) applications. Krizhevsky et al. [53] proposed AlexNet achieved high accuracy in the ImageNet contest 2012 with a proper approach reducing the overfitting issue. Overfitting is when neural networks are too aligned with training data resulting in poor generalisation. The success of AlexNet is a milestone. Since then, DNN has been applied to various fields. WaveNet [54] is able to generate raw audio waveforms with realistic human voices and is used to generate Google Assistant voices in US English.

Recurrent Neural Network (RNN) is a hypothesis that the neural network utilises the previous outputs as part of the input vector to the same neuron. Therefore, RNN can process the input of variable lengths and take into account historical information. Therefore, RNN outperforms other ANNs in tasks like natural language processing and time-series forecasting [55]. Long short term memory (LSTM) developed by Hochreiter and Schmidhuber [56] uses a "gate" concept in the neural networks to control the preservation of information so that LSTM can learn from historical information, reducing the gradient vanishing effect in the traditional RNN model [57]. Fig. 6 shows the structure of a LSTM cell. Successful attempts have been made in time-series data forecasting using RNN in different fields. Cao et al. [58] managed to make one-step ahead forecasting on decomposed financial time series using LSTM. There are similar research using LSTM to predict petroleum production [59], gold price [60]. Zhao et al. [61] applied a convolutional bi-directional LSTM (CBLSTM) model directly to raw sensory data from a machine health monitoring system of a computer numerical control (CNC) milling machine. The results show the proposed model can predict the actual tool wear. In civil SHM studies, monitoring data are normally presented in the time-series format. Therefore, the use of the recurrent neural network to analyse SHM data from long-span bridges is feasible.

These models discussed above are some of the commonly used ML algorithms. Also, different algorithms are specialised for specific types of tasks. Therefore, in the SHM studies, researchers have adopted these ML methods for different purposes, e.g. structural response predictions, damage status classifications. In the next section, several case studies regarding machine learning development in SHM studies will be discussed.

4. Case studies

This section surveys various ML applications in civil SHM studies. Different algorithms, including SVM, random forest, MLP, CNN and RNN, are used in the SHM problems in long-span bridges, dams, buildings and large wind turbines.

4.1. Multilayer perceptron (MLP)

Min et al. [62] introduces an impedance-based structural damage detection method using MLP. Wireless sensors first collect impedance signals. Then cross coefficients (CC) are used as the predefined damage index. Finally, an MLP model is trained on CC values to predict damage severity. This method successfully detects loose bolts and cracks on a steel bridge girder. Soyoz and Feng [63] trained an MLP model based on the modal parameters from vibration measurements to study ageing and long-term bridge structural parameters using data collected from an in-service bridge, Jamboree Road Over-crossing, California, USA. Mehrjoo et al. [64] used MLP to evaluate the damage status of joints on truss bridges. Mode shapes and the natural frequency of the bridge were used as the input to the model. The location and severity of damages can be accurately determined by the MLP. However, due to the limits of computation capabilities at the time of this research, the most complicated structure in this article is a truss bridge with 16 joints. Similar studies on the damage localisation on truss bridges using MLP are also performed by researchers [65,66]. Zhao et al. [67] used an MLP model to predict the mid-span girder deflections under the ambient temperature influence. Hakim and Razak [68] proved the MLP were capable of using the first five natural frequencies as input to determine the severity of the damage on the steel bridge girder. Pawar

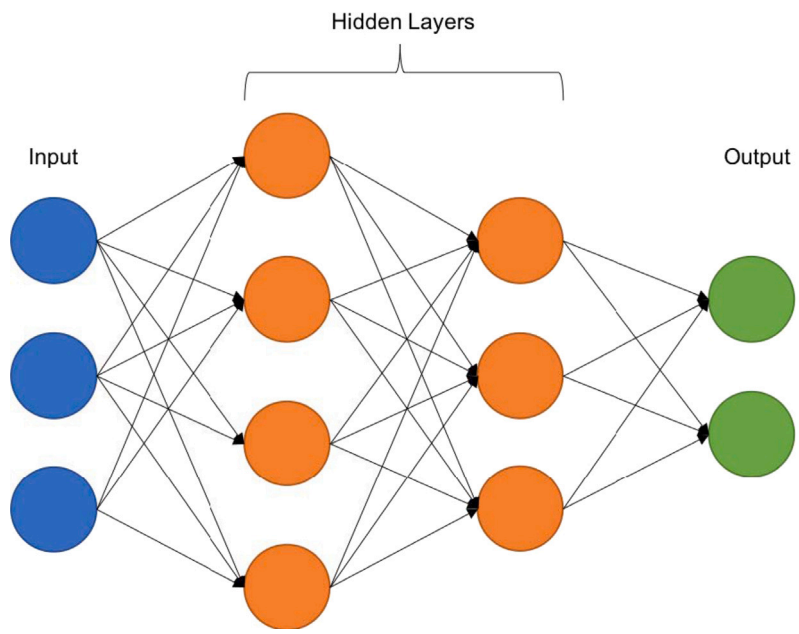


Fig. 5. Structure of the multilayer perceptron (MLP).

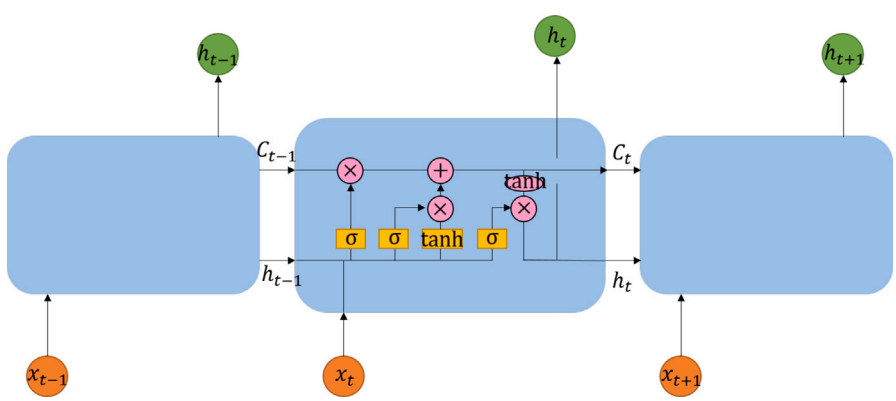


Fig. 6. Long-short term memory (LSTM).

Table 3
Other MLP applications on bridges SHM.

Algorithms	Structure	Preprocessing	Input	Output	Results	Reference
MLP	FE-model of a prestressed concrete bridge	Wavelet transform	Wavelet Normalised Index	Damage Occurrence	Mean probability = 95%	Taha et al. [73]
MLP	Lab test platform	PCA	Temperature-compensated frequencies	3 damage classes	Test set accuracy 99.8%	Concepcion and Ilagan [74]
ICA-ANN	Slab-on-girder bridge	Transform Formatting Cleaning, Feature engineering	Frequencies of first four modes	Damage Types	R-square: 0.99876	Gordan et al. [75]

et al. [69] used MLP to detect the damage location and size using a vector of Fourier coefficients of the mode shapes on a fixed–fixed beam. Other uses of MLP damage detection on beam-like structures are also reported [70–72]. Several other studies are listed in Table 3.

In summary, the use of MLP in SHM has been extensively researched over the past two decades, demonstrating its capability in damage detection primarily on simple structures like beams and truss systems. MLP has shown superior performance over other machine learning algorithms such as SVM and decision trees, although its application to in-service bridges remains limited. Most studies have utilised short-term monitoring data, often sourced from load tests or finite element simulations, and have predominantly focused on binary classifications like 'damaged' versus 'normal' structural status. However, research investigating MLP's effectiveness with long-term SHM monitoring data under real traffic conditions is scarce, due to the complexity of such data and MLP's limitations in processing long sequence data.

4.2. Convolutional neural network (CNN)

Due to benefits from the huge success of CNN in image processing, its application in structural health monitoring is extremely popular in recent years. Since structural damages can appear as cracks which can spread, enlarge and eventually lead to the collapse of the structure, detection and handling of cracks are vital [76]. Traditionally, cracks on structures are examined and detected by visual inspections by experts. However, this method is labour-demanding, expensive and subjective to human errors. With the development of image processing algorithms, automatic crack identification is now possible with machine learning methods, especially with CNN. Compared with MLP where each neuron is connected to all neurons in the following layer, as shown in Fig. 5, (dramatically increasing the trainable parameters and inefficiency) CNN utilises convolutional layers before fully connected layers where different filters slide across the image, allowing patterns to be recognised regardless of their locations in the image. Therefore, the trainable parameters are far less for a CNN model. As a result, a CNN model can have multiple convolutional layers to identify multiple features in an image with a smaller size of parameters to train and also achieve high accuracy. There have been multiple successful attempts using CNN in crack detection. Table 4 shows the studies analysed in this section.

4.2.1. Image-based damage detections

Cha et al. [77] used CNN to develop a model to detect concrete cracks from images. In their experiments, 277 images were manually labelled as cracked or intact to train the CNN model. The trained model achieved 97% accuracy. The shape and location of cracks can be identified using CNN, as shown in Fig. 7(b). Compared with the traditional Canny edge detection, Fig. 7(c) and Sobel edge detection, Fig. 7(d), the proposed CNN detection method can provide clearer crack information. The proposed method also benefits from simple data preparation needed. Modarres et al. [78] applied CNN to concrete surface images collected from a concrete bridge. Two sets of images are used to verify the feasibility of the CNN model, one clean set of well-controlled surfaces with different lighting conditions representing concrete in new bridges and one noisy set of textured concrete surfaces with unclear cracks representing in-service structures in a visual inspection. The results show that the proposed CNN outperforms other machine learning models, e.g., MLP, random forest, SVM with non-linear kernel in crack detection for both image sets. Zhang et al. [79] built a deep CNN model enabling the automatic detection of pavement cracks. Due to the low contrast between cracks, surroundings, and shadows, such a task can be challenging for traditional methods. The proposed deep CNN model achieved satisfactory performance compared with SVM and Boosting method. Similar researches are also performed by Xu et al. [80],Dung et al. [81],Chen and Jahanshahi [82],Alipour et al. [83]. Attempts have been made by using other popular deep CNN models for concrete crack detection, e.g. ResNet-50 [84], AlexNet [85], VGG-16, GoogLeNet [86].

Apart from directly using the CNN or deep CNN models, researchers incorporate different algorithms in the CNN layers to further improve accuracy or capability in complex crack detection problems. Xu et al. [80] introduced the Atrous Spatial Pyramid Pooling (ASPP) module in the CNN model. ASPP can avoid loss of information when the image is downsampled by traditional convolutional layers. This ASPP CNN model was trained on 4856 concrete surface images and the results outperform the conventional CNN model. Several studies have used Region-based CNN (RCNN) for detection. In this model, the output contains not only the classifications but also bounding boxes showing the region of interest (ROI), e.g. the locations of cracks. In this way, the size of cracks can be quantified by width and height. Suh and Cha [87] successfully used a faster R-CNN model to detect steel corrosion and concrete cracks automatically and achieved real-time monitoring. Similarly, Zhao et al. [85] proposed a comprehensive inspection method for bridge maintenance. First, an AlexNet was used to classify different types of bridges, e.g., suspension, cable-stayed bridges. Second, a Faster-RCNN is trained to identify different bridge components (e.g. tower, deck). Finally, concrete cracks are detected by a GoogLeNet. However, this research lacked discussions on the link among each of the three parts.

4.2.2. Time-series classification

Even though most studies on CNN application are image-based crack detection, a few researches apply CNN to time-series SHM monitoring data. Bao et al. [88] transformed time series signals from a long-span bridge SHM system into image vectors and manually labelled these images for different anomaly patterns, as shown in Fig. 8. Each subfigure represents data from a different channel. Then these images served as input to CNN to simulate the manual inspection process. The trained CNN can detect potential anomalies in large volumes of SHM data. Similarly, Tang et al. [89] split and visualised the raw time-series monitoring data in the time and frequency domain. Then, CNN is used to detect anomalies in these visualised data. The proposed methods are validated using acceleration data from an in-service long-span cable-stay bridge. Different strategies regarding time-series data visualisation are also proposed by others [90,91]. The implementation of 1D-CNN in some research [92,93] enables the direct use of time-series data without visualisation. Traditional CNN models take imagery data as input, so the input shape is *(height, width, depth)*, where *depth* is the colour channels in an image. However, the time-series SHM monitoring data is one-dimensional. Using 1D-CNN allows the convolutional kernel to slide in one direction only. Therefore, 1D-CNN can be used on SHM data without data visualisation and the 1D convolutional layer manages to extract features from the time-series data.

4.2.3. Summary

CNNs are a burgeoning research area in SHM, demonstrating superior accuracy in image-based damage detection over other ML algorithms. Different deep CNN models have been used to efficiently identify cracks and their locations, offering rapid and precise fault detection. However, acquiring the large volume of images required for training, predominantly a manual task, remains a challenge, despite attempts to automate the process using UAVs with cameras [94–96]. CNNs serve as a classifier for crack and anomaly detection, requiring labelled datasets for training. Given the vast data requirements of CNNs, the labelling process significantly adds to the workload.

4.3. Recurrent neural network (RNN)

For its ability to retain historical information from the input sequence, RNN is capable of tasks regarding natural language processing and time-series data analysis. There have been many attempts to use RNN for time-series forecasting. However, the application of RNN in civil SHM problems is rare compared to CNN. In an SHM system, data corruption is inevitable despite the advances in sensor hardware in recent years. Some typical data errors have been discussed in Section 3. There have been a few attempts using RNN to correct these errors. Jeong et al. [97] proposed a method using bidirectional recurrent neural networks (BRNN) for sensor data reconstruction. The assumption is that errors can be corrected by data sampled by other sensors nearby. The bidirectional feature used in this article allows the RNN to learn correlations between different sensor data from two directions, past to future and future to past. The results show the proposed BRNN method can reach more accurate estimations of the missing parts, compared to SVM, PCA, and RNN. One test example is given in Fig. 9 where missing data from 2 sensors are predicted using nearby sensors. However, one drawback is observed. The training time of BRNN is significantly higher than other models, in the meantime, the accuracy improvement is relatively low.

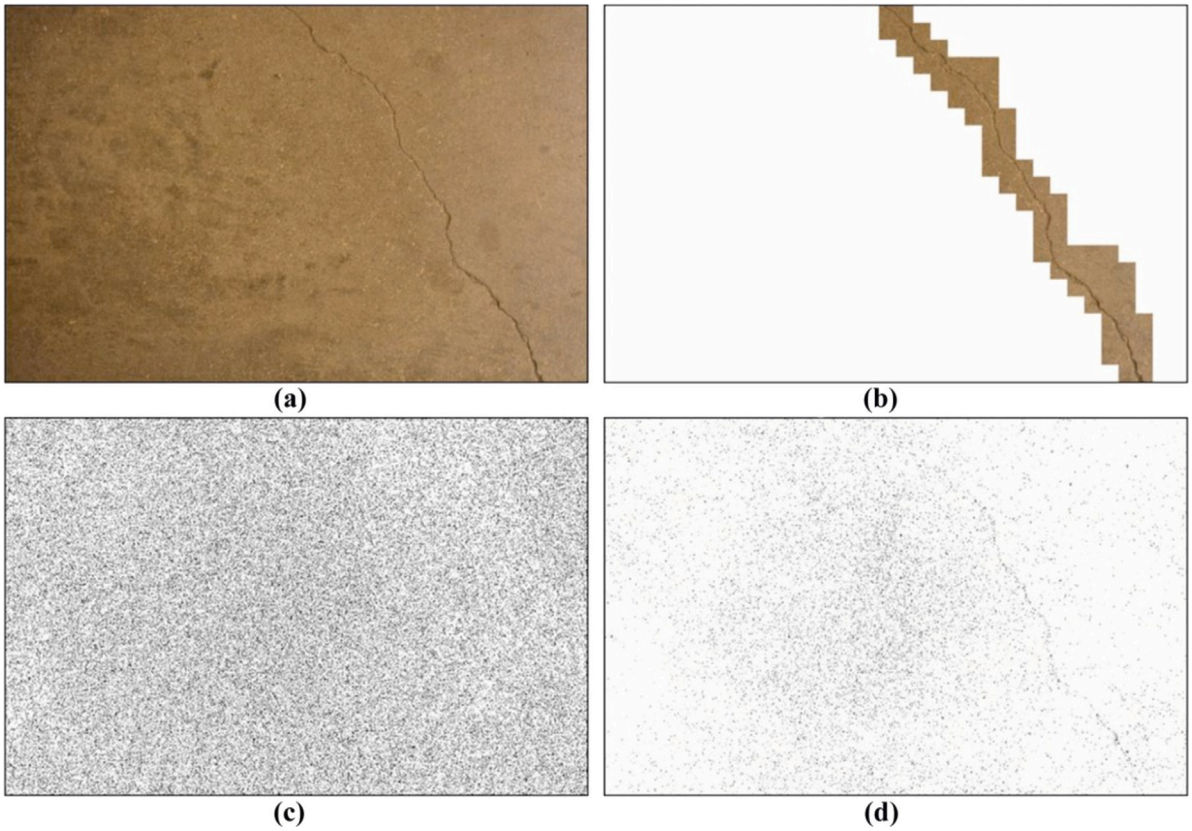


Fig. 7. Proposed CNN model compared with other detection methods [77].

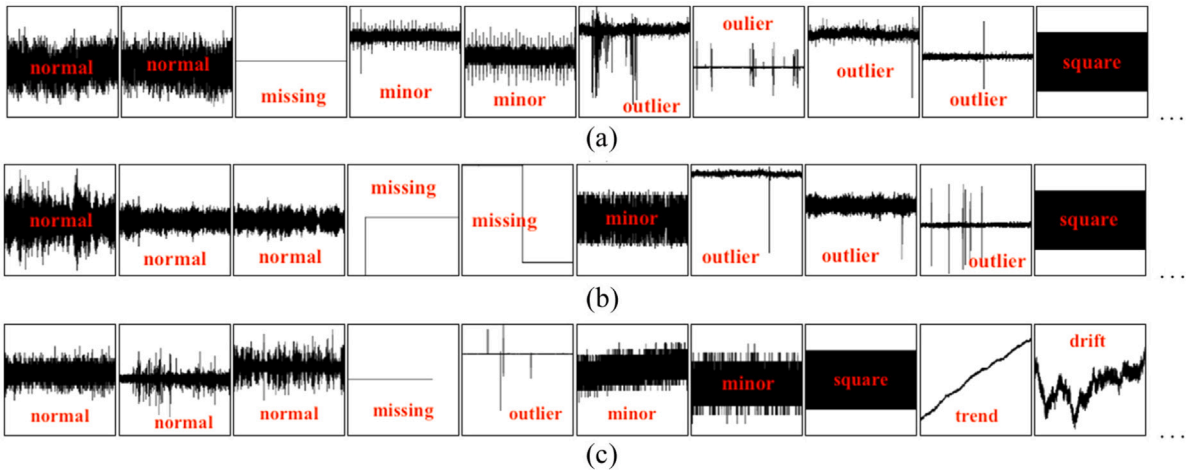


Fig. 8. Examples of data visualisation with anomalies labelled [88].

4.3.1. Regression analysis with RNN

For an in-service bridge, structural responses are affected by environmental and operational conditions. Therefore, the thermal-induced, traffic-induced and wind-induced bridge responses are popular research topics in the SHM studies. There have been several papers investigating the recurrent neural networks applications in the above topics. Yue et al. [98] successfully used a long short-term memory (LSTM) network [56], a kind of RNN, to predict thermal-induced deflections of a long-span bridge based on temperature data. The prediction results are shown in Fig. 10. LSTM is also used for the non-linear mapping between the dynamic displacement and the vehicle-induced strain [99]. Li et al. [100] investigated buffeting responses induced by wind effect using LSTM. In this research, the wind speed, direction and attack angle are preprocessed and used as input data. The vertical acceleration of the deck is used as the output. The prediction results are then compared with the buffeting responses calculated based on the Davenport–Scanlan's model and finite element method (FEM). LSTM predictions are more accurate than the conventional FEM model in both time and frequency domain. Similar research shows that the wind-induced bridge action can also be accurately predicted by LSTM [101]. Guo et al. [102] predicts girder deflections based on temperature, crack, humidity, strain and historic deflection data with a LSTM network. Despite the high accuracy recorded in this research, the proposed method only manage to predict deflections 3 steps further using a dataset of 8735 time steps for training and testing.

Table 4
CNN applications on bridges SHM.

Algorithms	Usual applications	Structure	Preprocessing	Input	Output	Results	Reference
CNN	Canny and Sobel edge detection	Concrete surface	Labelling	Images of concrete surface	Crack existence, locations	Accuracy: 97%	Cha et al. [77]
CNN	PCA, SVM, MLP, Random Forest	Concrete surface on bridges	Labelling, Add noise	Images of concrete surface	Damage types	Accuracy: 99.6%	Modarres et al. [78]
Deep ConvNet	SVM, Boosting	Pavement surface	Normalisation	Images of pavement surface	Crack existence	Precision: 0.8696 Recall: 0.9251	Zhang et al. [79]
ASPP-CNN	Resnet, Vgg	Bridge cracks	Reshape, labelling	Images of bridge cracks	Crack existence	Accuracy: 96.37%	Xu et al. [80]
FCN	(No discussed)	Concrete surface	Reshape, labelling	Images of concrete surface	Crack existence, locations	Precision: 90%	Dung et al. [81]
NB-CNN	LBP-SVM	Steel surface	Reshape, labelling	Video frames with noise (non-crack marks)	Crack bounding boxes	Hit rate: 98.3%	Chen and Jahanshahi [82]
CrackPix (CNN)	Edge detection	Concrete surface	Upsampling, resize, labelling	Images of concrete surface	Crack existence	Crack: 92% Non-crack: 99.9%	Alipour et al. [83]
Modified ResNet-50	ResNet-50, GoogLeNet	Concrete surface	Image processing (enhance, split, expand)	Accelerations	Defect existence, classification, Severity prediction	Defect: 90.4% Classification: 81% Severity: 78%	Kruachottikul et al. [84]
AlexNet (CNN), Fast R-CNN, GoogLeNet	(Not discussed)	Bridges	Labelling	Images of bridges	Bridge types, bridge components, Cracks existence & locations	AlexNet: 96% Fast RCNN: 90.45% GoogLeNet: 99%	Zhao et al. [85]
CNN	Deep MLP	A long-span bridge	Data visualisation,	Visualised time series data	7 types of anomalies	Overall accuracy: 93.5%	Tang et al. [89]
CNN	Deep MLP	A long-span cable-stayed bridge, China	Data visualisation, decomposed into 3 channels	Visualised & decomposed time series data	7 types of anomalies	Overall accuracy: 95.07%	Shajihan et al. [90]
GANomaly-CNN	(Not discussed)	A long-span cable-stayed bridge, China	Data visualisation, decomposed into 3 channels using FFT and GAF	Visualised & decomposed time series data	7 types of anomalies	Global accuracy: 98.2%	Liu et al. [91]
1D-CNN	1D-DCT	A suspension bridge, China	Labelling	Time series data	Binary classification	F-measure: 97.53%	Ni et al. [93]

The model is highly converged to the given dataset. Therefore, the generalisation on other data is questionable. There is one research on outliers detections using RNN. Mousavi and Gandomi [103] proposed a damage detection scheme using bidirectional LSTM model to learn the Mahalanobis distance, distance between a data point to distribution, of the natural frequency of the bridge when the bridge is under healthy state. Later, the model was verified by monitoring data with damage occurrence, where large errors were observed, indicating outliers existence. Different from other research where raw time-series data are used for training, the input data was preprocessed with Variational Mode Decomposition (VMD) to denoise and remove any seasonal patterns in the signals. Then, a Minimum Covariance Determinant (MCD) is used to obtain the Mahalanobis distance for training.

4.3.2. Damage classification

Damage classification using RNN is also proposed by different researchers. Yang et al. [104] used a Hierarchical CNN and gate-recurrent unit (GRU) method (HCG), an algorithm combining the feature of both convolution and recurrent layers, to perform damage detection on acceleration data from a bridge scale model. Concentrated force was applied to the main span creating cracks to represent structural damage. The prediction accuracy was compared with MLP, CNN and RNN. The proposed HCG model not only outperformed all other neural network methods on this damage detection task but also showed similar training time compared with other models. Sony et al. [105] divided acceleration data into multiple fixed-length windows to reduce the data dimensions and increase the size of training data sets. Each window is labelled with damage status. Then an LSTM model is used to classify acceleration responses. The model successfully predicted the damage class and probabilities of damage over the target bridge (damage localisation).

4.3.3. Summary

Currently, the applications of RNN in SHM study are quite recent compared to CNN and MLP. A vast majority of these researches focused on exploring the relationships between the environmental and operational conditions and the bridge structural responses. Such a relationship is normally complex and non-linear for in-service long span bridges. On the other hand, it is observed that time series data used in RNN models for regression analysis do not require sophisticated data preprocessing. In several cases, raw data were directly used with only standard normalisation [98,99,101,102]. This can drastically reduce the complexity of the data preparation, eliminate the time-consuming and labour-intensive labelling process compared with CNN applications. Table 5 lists the studies investigated in this section.

4.4. Other ML methods

Apart from the huge number of attempts at using neural networks for damage detection and condition assessment on bridges—other machine learning models, e.g. SVM and random forest, are evaluated for various topics. The studies in this section are summarised in Table 6.

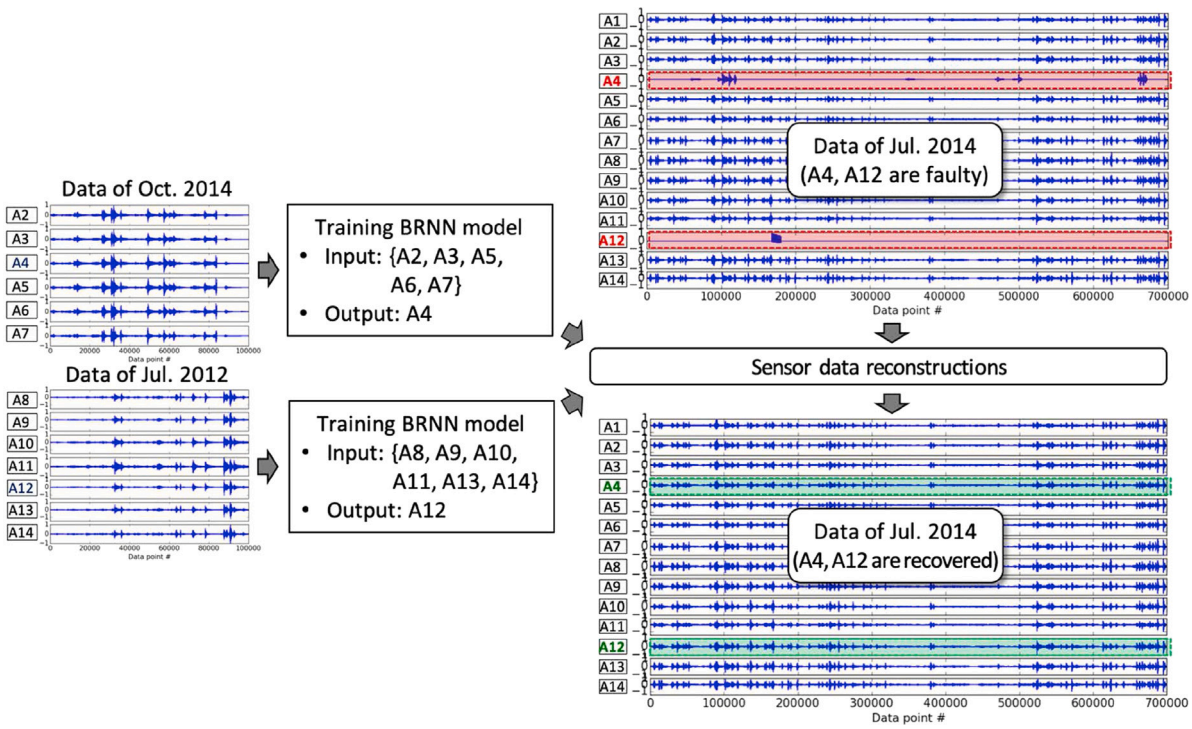


Fig. 9. Missing data recovery with BRNN [97].

Table 5
RNN applications on bridges SHM.

Algorithms	Usual application	Structure	Preprocessing	Input	Output	Results	Reference
BRNN	PCA, MLP	Telegraph Road Bridge, Michigan, US	Normalisation, Scaling	Vibration responses from nearby sensors of the target	Missing Vibration responses of the target	(Numerical metrics not provided)	Jeong et al. [97]
LSTM	Linear regression	A steel truss girder cable-stayed bridge over the Yangtze River, China	Normalisation	Temperature or Deflection+ Temperature	Displacement	Average error: <0.5mm	Yue et al. [98]
LSTM	SVM with Gaussian kernel	A 25 m box-girder bridge, China	Normalisation	Temperature	Strain or Displacement	(Numerical metrics not provided)	Zhao et al. [99]
LSTM	Davenport–Scanlan's buffeting force model	A suspension bridge, China	Normalisation, Feature extraction	Wind speed, Wind direction	Buffeting responses	(Numerical metrics not provided)	Li et al. [100]
LSTM	Quasi-steady theory	A unit-length section of a bridge deck	Normalisation	Deck displacement	Aerodynamics forces and responses	(Numerical metrics not provided)	Li et al. [101]
LSTM	(Not discussed)	A bridge in Hubei Province, China	Normalisation	Humidity, Temperature, Crack, Strain, Deflection	Deflection	95% predictions with error <1mm	Guo et al. [102]
BiLSTM	PCA-based method	Z24 Bridge, Switzerland	VMD, FastMCD	Mahalanobis distances of vibration data	Mahalanobis distances of vibration data	(Numerical metrics not provided)	Mousavi and Gandomi [103]
CNN-GRU	CNN or RNN only	A scale model of a bridge	Normalisation, Labelling	Vibration data	4 damage states	Accuracy: 0.945-0.949	Yang et al. [104]
LSTM	1D-CNN	Z24 Bridge, Switzerland	Normalisation, Labelling	Vibration data	3-4 damage classes	Accuracy 94%	Sony et al. [105]

4.4.1. Transformer

The Transformer is a deep-learning model that is based on the attention mechanism [106]. It stands out for its reduced training time requirements compared to earlier recurrent neural network architectures, including LSTM. Transformer has gained great success in both natural language processing [107] and image processing [108,109].

Although great success has been observed for the transformer model, its application in the bridge SHM is very rare. Yang et al. [110] trained an encoder-only transformer model with multi-dimensional output to derive a complex dynamic spatiotemporal mapping across various measurement points from an undamaged structure, establishing a benchmark model. Data from eight sensors were used to train the

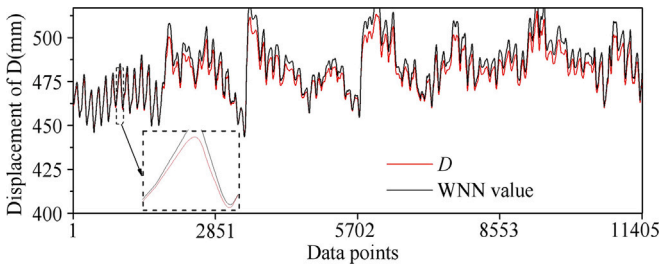


Fig. 10. Displacement predictions with LSTM [98].

model. The results were evaluated on both one-dimensional and multi-dimensional outputs and showed effective establishment of inter-sensor relationships.

4.4.2. Support vector machine

SVM is used to determine damage location and severity. For example, He and Yan [111] introduced the wavelet basis as the kernel function of SVM. The proposed wavelet support vector machine was then deployed to detect damages under ambient excitation. Pan et al. [112] used a radial basis function (RBF) kernel SVM to perform classification on damaged and undamaged states of a cable-stayed bridge. Compared with other SVM applications, different data preprocessing methods are investigated in this research. It was found that wavelet transform has a higher accuracy in noisy data, whilst Teager–Huang transform (THT) performs better on regular signals with limited noise. Li et al. [113] used SVM on the impact-echo data, an acoustic inspection method, collected from a road bridge in New York to classify deck damage states. In this research, the data are preprocessed with Fast Fourier Transforms (FFT) and Wavelet Transforms (WT). An attempt was made to investigate the relationship between fatigue damage of hangers on a suspension bridge and live traffic loads monitored by the weigh-in-motion (WIM) system [114]. The proposed SVM model accurately predict the fatigue damage on the target hanger using traffic volumes and equivalent traffic loads. Furthermore, it was found that the increasing traffic loads can greatly accelerate the fatigue damage on hangers. Image-based cable defects identification using SVM was also reported [115].

4.4.3. Random forest

Random forest is another machine learning algorithm that is widely used in SHM studies. Lei et al. [116] proposed the random forest method to predict the damage state for regional bridges under seismic conditions. The model achieved over 90% accuracy on damage classification. Random forest is also used for imputation to reconstruct the missing monitoring data from the Forth Road Bridge [32].

4.4.4. Clustering

Unsupervised learning methods are also employed in damage detection studies due to the large amount of raw SHM monitoring data and labour-intensive labelling process required for any supervised learning methods. Diez et al. [117] used K-means based cluster on joints vibration data from the famous Sydney Harbour Bridge. The proposed approach managed to group vibration data showing similar behaviour. Therefore, the vibrations under undamaged and damaged states are grouped into different clusters, allowing damaged joints to be detected. A similar research used the K-medoids clustering method to perform damage detection on the Z24 Bridge vibration data [118], on the PI-57 Bridge, France [119].

In this section, several other machine learning algorithms used in SHM studies are discussed. SVM is very popular for its non-linear kernel function on complex classifications. Transformer is a relative new method and are effective for multidimensional outputs.

5. Discussion

Above is a detailed review of the ML applications in civil structural health monitoring studies on concrete and steel bridges. Analyses are presented on different algorithms, their advantages and disadvantages. Tables 3, 4, 5, 6 present summaries of different machine learning studies in the bridge SHM field. Since the traditional FE-based assessment model is difficult to develop for real bridges, data-driven statistical models are popular to use for SHM data analysis. Furthermore, due to the complexity and the volume of the monitoring data, conventional models can have a poor generalisation ability on new datasets. Therefore, using machine learning algorithms for SHM problems can improve long-term monitoring reliability and potentially reduce the maintenance cost of in-service bridges as it can detect structural changes based on data, independent from visual inspections. In general, machine learning methods are applied in SHM for condition assessment, e.g. different damage states classifications, and damage detections, e.g. image-based concrete surface crack detections.

For the very basic ANNs, the MLP, is successful in certain areas, e.g., mode shape analysis. It is more accurate in predictions compared with other methods. However, due to the simplicity of the neurons used in MLP, it is only suitable for classifications for damage to simple structures in laboratory environments. On the other hand, MLP only uses fully connected layers. The trainable parameters will increase drastically when more neurons or hidden layers are used in one network, limiting its use in deep learning models. Currently, the SHM systems on in-service bridges not only record bridge structural responses but also their environmental and operational data, e.g. temperature and traffic information. The complex monitoring data help engineers evaluate the bridge states more accurately. However, it is difficult for MLP to learn trends and patterns from these complicated multi-channel data.

Convolutional Neural Network (CNN) is the most popular neural networks model in SHM studies in recent years. As a deep learning model, it is capable of processing complicated multi-channel training data. The convolution layers not only extract features from the input dataset but also reduce its dimension. As a result, the scale of the trainable parameters is controllable and the development of a deep neural network with convolutional layers is possible. In the SHM field, the most successful application is image-based crack detection. In certain damage detection tasks, it can achieve an overall accuracy of 97%. The automation of concrete crack detection becomes possible with this ML method. To increase the robustness of the detection model, some researchers manually add noise or distortion to the training images to simulate different shooting conditions [103]. One other feature of image-based damage detection methods is that it is a universal solution not only applicable to bridges but also other civil infrastructures, e.g. dams, high-rise buildings. Furthermore, from Table 4, the data preprocessing is relatively straightforward, normally only involving labelling and reshaping. However, it is recommended for future research that the acquisition procedure of trainable images should be further improved. Currently, most of the reviewed studies use photos manually taken by DSLR cameras or smartphones. Due to the nature of neural networks, a great number of images are required in training for better accuracy. Therefore, automatic image acquisition methods should be considered in future works. On the other hand, using UAVs in image acquisition allows the inaccessible elements on long-span bridges to be inspected. However, adverse weather conditions, such as high wind speeds, can introduce motion blur issues to images captured by UAVs, thereby reducing the accuracy of damage detection. Consequently, image preprocessing is necessary to mitigate these blur effects [120]. For its exceptional performance in image processing, some researchers attempted to visualise the time-series SHM data which then can serve as the input data to CNN. The results also indicate successful detections of structural damages. Therefore, in the data pre-processing stage, the manual labelling of the training data is mandatory. As a result, one notable drawback of CNN is the requirement for data labelling which

Table 6
Other MLP applications on bridges SHM.

Algorithms	Usual application	Structure	Preprocessing	Input	Output	Results	Reference
SVM	FE model-based fatigue analysis	Nanxi Suspension Bridge, China	Input: traffic volumes and traffic weights Output: fatigue damage generated by FEM using WIM data	Bridge-site traffic loading parameters	Fatigue damage of hangers	R-square: 0.90–0.99	Deng et al. [114]
PSO-SVM	Zernike moments, Gabor filter, Wavelet	Cables on long-span bridges	Texture extracted by GLCM, Manual labelling	Grey-scale cable images	4 damage classes	Accuracy: 96.25%	Bu et al. [115]
SVM	Visual inspections, GPR, PSA	Flushing 149st Bridge, New York, US	FFT, WT	Impact-echo data	5 damage types	Detected flaws: 13.3% closer to 12.7% from empirical analysis	Li et al. [113]
SVM	Physics-based methods	Manavgat Cable-stayed Bridge, Turkey	WT, HHT, THT	Vibration data	15 damage types	Accuracy: 0.9688–0.9900 Error: 0%-5.33%	Pan et al. [112]
Random Forest	Traditional methods using material stiffness and boundary conditions	Nanli River Bridge, China	Labelling	Ground motion data, Bridge dynamic parameters, Bridge design parameters	Seismic damage states	Accuracy: 90%	Lei et al. [116]
Random Forest	Linear regression	Forth Road Bridge, UK	Outlier removal	Strain, Temperature	Missing strain data	Accuracy: 92%	Xu et al. [32]
K-means Clustering		Sydney Harbour Bridge, Australia	k-Nearest Neighbours(kNN), FFT	Vibration data	Clusters	Clusters	Diez et al. [117]
K-medoids Clustering		Z24 Bridge, Switzerland		Vibration data	Clusters	Clusters	Sarmadi et al. [118]
K-medoids Clustering		PI-57 Bridge, France	Discrete Fourier Transform	Vibration data	Clusters	Clusters	de Almeida Cardoso et al. [119]

can be both time-consuming and monotonous. Nevertheless, CNN is still an active research area in SHM studies and is recommended for future exploration, especially in its application to the time-series monitoring data.

The application of the Recurrent Neural Network (RNN) is relatively new to SHM studies, especially in regression analyses on SHM data. Most studies focus on exploring the relationship between the bridge environmental, operational conditions, e.g. thermal effect, wind load, traffic load, and the structural responses. Different from 1D-CNN applications on time series data, RNN can actually predict the future structural responses, e.g. accelerations, deflections. First, the prediction of the exact structural response is much more difficult than classifications. On the other hand, Long-short term memory (LSTM), one type of RNN, is more easily affected by the overfitting problem than other neural network models [121]. Therefore, regression predictions with RNN can be a challenging task. However, its advantage is eminent as the predicted structural responses give engineers direct insight into the structure. Unlike the classifications, only the final damage status is given, further probability models can be applied to the predicted responses so that engineers can determine whether specific structure components are close to damage. Furthermore, from Table 5, it can be observed that most of the RNN applications in bridge SHM only require the minimum data preprocessing compared with CNN or MLP models. As a result, RNN models can be versatile to different datasets. It is recommended to continue exploring different regression analysis approaches for SHM data.

As for other machine learning methods, the applications of SVM and random forests are limited by the requirements of complicated data preprocessing steps, as shown in Table 6. K-means clustering is the most popular unsupervised method to group unlabelled data. Therefore, researchers are encouraged to seek possibilities to replace manual labelling with K-means clustering in the data preparation step for other supervised methods. Transformer is a new and more efficient deep learning model in natural language processing and image processing. It has great potential in regression analysis in multidimensional output and requires further applications in bridge SHM.

6. Conclusions

Machine learning applications to concrete and steel bridge structural health monitoring system have become a very popular research topic in recent years. There have been a great number of researchers in the field exploring the significance of ML in bridge condition assessment and damage detection. In this article, we reviewed different machine learning algorithms and their applications to in-service concrete and steel bridges. We compared the feasibility of different algorithms for different bridge damage detection tasks, some caused by material deteriorations. Detailed tables are provided for clear comparisons of different machine learning methods used in bridge SHM.

Based on the detailed review, we can draw the following conclusions:

- Data preprocessing is essential for any ML applications. The strategies vary depending upon different types of SHM data and ML algorithms used.
- The basic ANN model, MLP, is suitable for simple numerical or experimental structures instead of full scale in-service bridges. Also, its ability to process multi-channel data is limited
- CNN applications on image-based damage detection and condition assessments are very successful, making it easy to detect damages caused by steel corrosions and concrete cracks.
- 1D-CNN can be directly used on time series data for damage classifications
- RNN is the new trend in SHM studies for its ability to make regression predictions on structural responses. Furthermore, RNN are used to map the non-linear relations between the structural responses and bridge environmental & operational conditions. On the other hand, data preprocessing for time series RNN applications is relatively basic.
- SVM and random forests receives less attention compared to neural networks. It is used for damage classifications. However, the data preprocessing is much more complicated.

- K-means clustering is the most widely used unsupervised learning methods in SHM. It shows success in automatically grouping data with no labels.

Machine learning in bridge structural health monitoring is a very active research field. For future developments, researchers are encouraged to make efforts in the following directions:

- It is highly recommended for researchers to explore the application of the Transformer model in the analysis of SHM data. At present, the Transformer model has achieved significant success in the realms of image processing and natural language processing. However, its application in SHM remains relatively unexplored. The model has demonstrated superior accuracy coupled with expedited training. Consequently, the application of the Transformer model to SHM data is anticipated to significantly enhance generalisation accuracy, particularly for complex, long-term monitoring data.
- Despite the abundance of data from SHM systems, the scarcity of data corresponding to various damage scenarios for in-service bridges limits the training dataset for ML. Currently, most ML methods learn solely from SHM data, lacking understanding of the underlying system. To address this, we propose Physics-Informed Machine Learning models that incorporate physical laws, such as using physical models for simulated data or embedding physical laws in loss functions. Future ML research in SHM should not just explore the feasibility of existing ML models, but also encourage modifications to ML algorithms based on SHM data.
- Image-based damage detection utilising neural networks has been extensively researched. However, to facilitate broader industrial application of existing research, standardisation is imperative. For instance, a uniform method for image acquisition is required to ensure consistent input data. A standardised pre-processing workflow could also contribute to reducing the time spent on data preparation. Consequently, it is highly recommended for researchers to investigate strategies for standardising the analysis of SHM data using machine learning.

CRediT authorship contribution statement

Donghui Xu: Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Xiang Xu:** Conceptualization, Investigation, Writing – review & editing. **Michael C. Forde:** Conceptualization, Funding acquisition, Investigation, Resources, Supervision, Writing – review & editing. **Antonio Caballero:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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